Islands of Reach

A censorship-resistant federated forum for national internet blackouts

Signed public discussion across the IP paths that still exist

NSysS 2026

Cox’s Bazar, Bangladesh

When the outside internet disappears

A public forum normally depends on distant platforms, identity providers and globally reachable servers.

1 **Traffic is throttled**

Messages become slow and unreliable

2 **Services are blocked**

Circumvention still needs a path outside

3 **International transit is cut**

Domestic networks may survive as separate islands

The design question

Can independent domestic servers keep a readable, verifiable forum working across the remaining ISP links?

Scope: surviving IP connectivity. This is not mesh or off-grid networking.

Sources: OONI (2025); Bischof et al. (2025); Telex (USENIX Security 2011).

SIMPLIFIED ABSTRACT

The solution in one minute

Independent servers

Communities can run their own nodes and choose their federation peers.

Signed objects

A post is valid because its author signed it, not because one server approved it.

Portable evidence

A client stores the node receipt and can copy it to independent audit-log servers.

Bridged ISP islands

A trusted multi-homed node relays verified objects across surviving domestic paths.

If no path remains between two components, each side keeps working locally and queues data. Crossing resumes only when a bridge can reach both sides.

Resilience is a ladder, not a single claim

L0	**Ordinary reach**	Global paths are available
L1	**National partition**	International transit is withdrawn
L2	**ISP island**	Domestic peering is also cut
L3	**Bridged islands**	A multi-homed operator joins two islands

What is measured

The prototype exercises L0 through L3. It does not promise delivery when every physical link is gone.

Related systems solve parts of the problem

System family	What it gives	Gap for this setting
Publius, Freenet, Tangler	Host takedown and deletion	Readers still need a route to replicas
Tor, Snowflake, Telex	Blocking and circumvention	They require some path beyond the censor
Netnews, ActivityPub, Matrix	Independent servers and public discussion	No scope-aware bridge with this evidence path
Nostr, SSB, AT Protocol	Signed or content-addressed objects	Different relay, identity or partition assumptions

The contribution is the combination: server federation, no-inbound participation, gossiped evidence and operation across L0–L3.

Selected sources: Waldman et al. (2000, 2001); Clarke et al. (2001); RFC 5537; RFC 9162; W3C ActivityPub; Tarr et al. (2019).

Step 1: the author signs the post

Write

Post, vote, moderation action or revocation

❯

Encode

One canonical byte representation

❯

Sign

Ed25519 signature covers those bytes

❯

Identify

SHA-256 content ID comes from the same bytes

Every feature enters through the same write endpoint.

A relay cannot alter or re-encode the signed bytes.

No email address, phone number or password is required for a pseudonym.

Methods: RFC 8032 Ed25519; deterministic protobuf profile; cross-language conformance vectors.

Step 2: every write passes the same 19 checks

1–12	13–15	16–17	18–19
Reject before storage size, canonical parse, domain, clock, signature, certificate, dedupe, replay	**Admission and meaning** anti-abuse, authorisation, body validation	**One transaction** apply projection + append Merkle witness	**Acknowledge and relay** signed receipt + federation fanout

Federated objects repeat every applicable check. Only the origin-specific payment at step 13 is not charged twice.

- Steps 1–12 cannot turn invalid input into database write load.
- Steps 16–17 prevent a visible post from existing without a matching log entry.

Step 3: independent servers exchange verified objects

Verify again

Trust changes quota, never validity.

Keep exact bytes

A peer cannot repair malformed encoding.

Work behind CGNAT

Outbound drains and requested backfill need no inbound address.

Recover after gaps

Durable queues and resumable backfill continue after a path returns.

Step 4: the audit-log server preserves the receipt

1. Client
queue signed request

2. Node
validate and commit

3. Receipt
tree head + proof

4. Certificate
exact request + receipt
save locally first

ALS A
Recompute IDs, verify signatures and inclusion proof, append hash chain.

ALS B
Run the same checks and keep an independent copy.

Issuing node
/status reports online, hidden, deleted or wrong-server status.

The ALS proves prior acknowledgement. It does not moderate, report live state, guarantee storage or recover a body after every copy is gone.

Direct-message certificates stay local to avoid creating a timestamped social graph.

Step 5: visible moderation, local identity

Moderation

A moderator signs an additive action. Clients can show the policy and its reason.

Deletion

A signed tombstone records deletion. Receipts still prove prior acknowledgement.

Choice

A user may move to another server with a different policy and still read federated content.

One device seed

- Community A key
- Community B key
- Community C key

Anti-abuse without civil identity

Proof of work, credits, blind credentials and epoch nullifiers price posting. They limit rate, not the number of identities.

Honest limit: network observers can still correlate timing and traffic volume.

Sources: Dwork and Naor (1993); Chaum (1983); RFC 9576; UN Special Rapporteur on anonymity (2015).

METHOD STEP 6

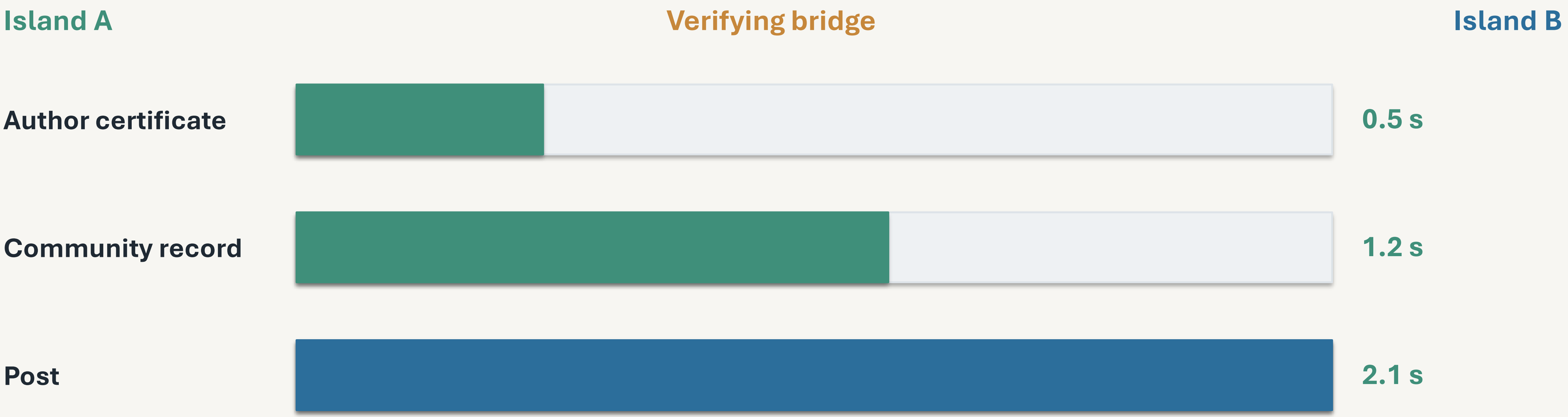
Bridge two ISP islands

- Trusted on both sides
- Outbound sockets bound to the correct source interface
- Quota counted across the uplink pair
- Half the grant reserved so bulk forum traffic cannot starve urgent classes

A bridge is a relay policy after verification, not a twentieth validation step.

MEASURED CROSSING

After the cut, dependencies cross in order



First deployed run

407.6 s

Cause

one blackholed peer blocked a serial drain

After the fix

2.1 s

Concurrent per-peer drains with deadlines

What the design resists, and what it does not

Resisted or bounded	Not solved by this design
• Post modification: author signature and content ID • Deletion after acknowledgement: client and ALS retain receipts • Opaque moderation: signed actions and visible local policy • Invalid-input write amplification: no-write validation prefix • One failed ISP path: ranked paths, bridge and durable retry • Peer equivocation: signed, gossiped tree heads	• No traffic-analysis or stylometry resistance • No safety after endpoint or signing-key compromise • No guarantee if every link or independent copy disappears • No identity-based Sybil prevention; posting cost only bounds rate • No fair or universal moderation policy • No guarantee against a compromised majority of peers or auditors

Condition for crossing: at least one authorised bridge must still reach both components.

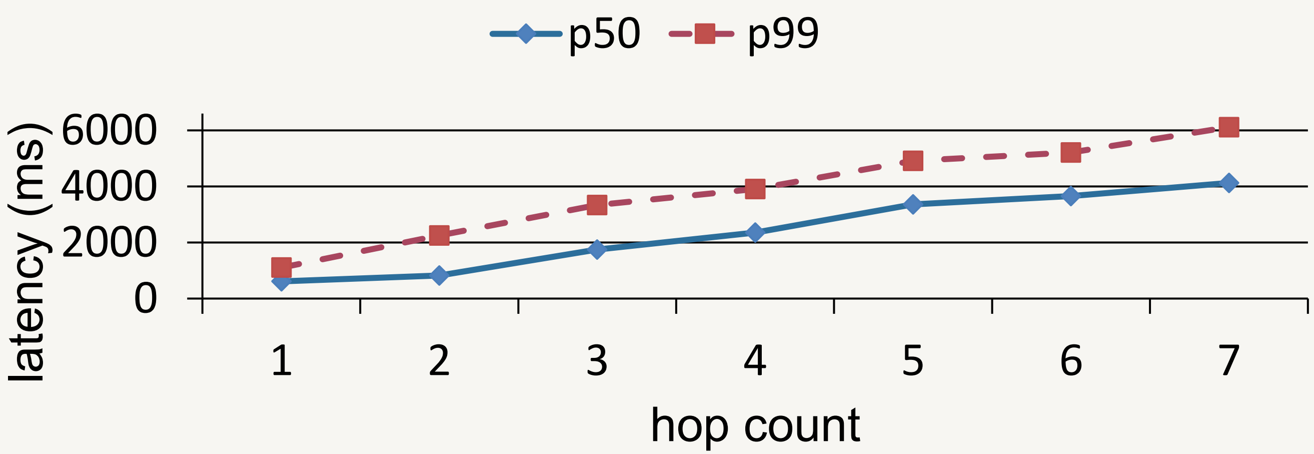
Prototype results

16 / 16
canonical vectors agree
TypeScript, Rust, Python

0 loss
200 posts reached hop 7
median 4.125 s

2.1 s
cut-path post rendered
after dependencies crossed

730 req/s
invalid flood rejected
zero database writes



Small-node footprint

62 MiB idle node
233 MiB under crossing
≈384 MiB node + database + cache

Wire size

155 B check-in
220 B forum post
243 B Bangla broadcast

TAKEAWAY

What this project establishes

A forum can continue inside surviving domestic IP components without trusting one platform or one administrator.

Local continuity

Each reachable island can publish, read and moderate locally.

Verifiable exchange

Signed bytes, receipts and audit copies make tampering visible.

Conditional crossing

A trusted bridge reconnects islands only while it can reach both.

Next evidence needed: a real multi-ISP field trial with separate clocks, independent auditors and measured censor interference.

No mesh claim. No absolute anonymity claim. No delivery claim when every path is gone.